

KH26-CSE-01 – AI-Driven Emergency Resource Allocation and Response Coordination System

Background

During floods, cyclones, fires and other emergencies, several locations may simultaneously request food, drinking water, medicines, rescue teams and temporary shelters. Available resources are limited, so processing requests only in the order received may delay support to highly affected locations.

Problem Statement

Develop an intelligent platform that receives emergency requests from multiple locations and calculates a dynamic priority score. The calculation must consider affected population, injuries, children, senior citizens, food and water availability, accessibility, distance and waiting time. The system should recommend how available resources should be distributed and display the suggested service order. Administrators must be able to update resource stocks and monitor the status of every request. The recommendations should include a simple explanation of why a location received a particular priority.

Mandatory Requirements

- Emergency officer and administrator roles
- Dynamic priority calculation
- Resource-allocation recommendations
- Critical-location alerts
- Request status tracking
- Explainable priority decisions

Expected Prototype

A responsive application demonstrating the complete process from emergency-request submission to resource allocation using simulated disaster data.

KH26-CSE-02 – AI-Based Multilingual Public Grievance Resolution Platform

Background

Government departments and public institutions receive complaints through different channels and languages. Manual classification, forwarding and tracking often lead to delays, duplicate complaints and unresolved grievances.

Problem Statement

Develop an AI-enabled platform that accepts grievances through text or voice in multiple Indian languages. The system should identify the grievance category, urgency, responsible department and possible duplicate complaints. It should generate a short summary, assign the complaint to the appropriate authority and provide a tracking facility to the citizen.

Administrators should be able to monitor pending, resolved and repeatedly reported issues through a dashboard. The system must also identify complaints that require immediate attention.

Mandatory Requirements

- Text or voice-based grievance submission
- Multilingual input support
- Automatic complaint classification
- Duplicate-grievance detection
- Department assignment and tracking
- Administrative analytics dashboard

Expected Prototype

A working grievance-management portal demonstrating submission, AI-based classification, assignment and status tracking.

KH26-CSE-03 – Intelligent Healthcare Referral and Hospital Capacity Management System

Background

Patients are often referred to hospitals without knowing whether the required specialist, bed, equipment or emergency facility is available. This can increase waiting time and delay critical medical attention.

Problem Statement

Develop an intelligent healthcare coordination platform that recommends an appropriate hospital based on patient requirements, urgency, distance and available facilities. Hospitals should be able to update bed availability, specialists, emergency services and essential equipment. The system should prioritize critical cases and suggest alternative hospitals when the nearest facility lacks the required resources. It should support patient registration, referral tracking and hospital-capacity visualization. The application must clearly state that its recommendation supports, but does not replace, professional medical decisions.

Mandatory Requirements

- Patient and hospital user roles
- Hospital-capacity management
- AI-based referral recommendations

- Emergency-case prioritization
- Alternative hospital suggestions
- Referral and admission tracking

Expected Prototype

A web or mobile application demonstrating patient referral to suitable hospitals using simulated capacity and location information.

KH26-CSE-04 – AI-Powered Digital Misinformation and Deepfake Verification Platform

Background

False news, manipulated images and synthetic media can spread rapidly through social platforms. Ordinary users may find it difficult to determine whether digital content is authentic or misleading.

Problem Statement

Develop an AI-assisted verification platform that analyses news text, images or short media samples for possible misinformation or manipulation. The system should examine the content source, writing patterns, visual inconsistencies and available supporting evidence. It should generate a credibility score and highlight the reasons behind the assessment. Users should be able to submit suspicious content and review previously analysed cases. An administrator dashboard should display common misinformation categories and emerging patterns.

Mandatory Requirements

- Text and image-content submission
- Credibility or risk scoring
- Suspicious-element highlighting
- Source and evidence comparison
- Explainable analysis
- Misinformation trend dashboard

Expected Prototype

A functional platform that analyses sample text and images and produces an explainable credibility report.

KH26-CSE-05 – AI-Based Public Distribution and Essential Supply Monitoring System

Background

Essential commodities such as food grains, medicines and relief materials may pass through several distribution points before reaching beneficiaries. Lack of visibility can result in shortages, delayed delivery, unequal allocation or abnormal stock movement.

Problem Statement

Develop an intelligent supply-monitoring platform that tracks stock from a central warehouse to local distribution centres. The system should analyse stock levels, demand, delivery delays and beneficiary requirements to identify possible shortages or unusual transactions. It should forecast demand and recommend stock redistribution between centres. Administrators must receive alerts when inventory falls below a critical level or when recorded distribution patterns appear abnormal. Citizens or beneficiaries should be able to check availability and delivery status.

Mandatory Requirements

- Warehouse and distribution-centre management
- Stock and delivery tracking
- Demand forecasting
- Shortage and anomaly alerts
- Stock-redistribution recommendations
- Beneficiary information interface

Expected Prototype

A dashboard demonstrating stock movement, shortage prediction and redistribution recommendations using simulated supply-chain data.

KH26-CSE-06 – Explainable Student Risk Identification and Academic Intervention System

Background

Students facing academic, attendance or engagement difficulties are often identified only after their performance has significantly declined. Institutions require an early-support mechanism rather than a system that merely labels students as weak.

Problem Statement

Develop an AI-supported platform that analyses attendance, assessment performance, assignment completion and learning participation to identify students who may require assistance. The system should explain the factors contributing to the risk indication and recommend suitable academic interventions. Faculty members should be able to record mentoring actions and monitor subsequent improvement. Students should receive

constructive recommendations without being shown harmful or discouraging labels. The platform must prioritize transparency, privacy and responsible use of student data.

Mandatory Requirements

- Student and faculty dashboards
- Early-risk identification
- Explainable contributing factors
- Personalized intervention suggestions
- Mentoring-action tracking
- Privacy-aware data handling

Expected Prototype

A working academic-support platform demonstrating student analysis, intervention recommendations and progress monitoring using sample data.

KH26-CSE-07 – AI-Driven Urban Mobility and Multimodal Route Planning System

Background

Commuters may use buses, shared vehicles, walking routes and private transport during a single journey. Existing navigation systems may not adequately consider crowding, delays, affordability, accessibility and environmental impact together.

Problem Statement

Develop an intelligent route-planning platform that recommends travel options using multiple transport modes. The recommendation should consider travel time, estimated cost, congestion, accessibility requirements and environmental impact. The system should adjust suggestions when a route becomes delayed or unavailable. Users should be able to select preferences such as minimum cost, shortest time or low-carbon travel. An administrative dashboard should display frequently congested routes and mobility demand patterns.

Mandatory Requirements

- Multimodal route recommendations
- User preference selection
- Dynamic route modification
- Cost and travel-time comparison
- Accessibility-aware suggestions
- Traffic and demand analytics

Expected Prototype

A map-based application demonstrating route recommendations and dynamic rerouting using simulated transport and traffic data.

KH26-CSE-08 – AI-Powered Government Document Intelligence and Workflow Automation System

Background

Government offices and educational institutions process large numbers of applications, certificates, letters and supporting documents. Manual verification and routing can cause delays, missing information and inconsistent processing.

Problem Statement

Develop an intelligent document-processing platform that accepts uploaded forms and supporting documents. The system should extract important information, identify missing fields, classify the application type and route it to the appropriate department. It should generate a concise summary for the reviewing officer and flag possible inconsistencies or duplicate submissions. Applicants must be able to track the processing status, while administrators should monitor pending and delayed applications. Human approval must remain compulsory for important decisions.

Mandatory Requirements

- Document upload and information extraction
- Missing-field identification
- Application classification and routing
- Duplicate or inconsistency alerts
- Applicant status tracking
- Human approval workflow

Expected Prototype

A working document-management system demonstrating upload, extraction, validation, routing and approval of sample applications.

KH26-CSE-09 – AI-Based Skill-Gap Analysis and Placement Readiness Ecosystem

Background

Students often prepare for placements without understanding the specific skills required by different job roles. Generic recommendations may not reflect their current abilities, academic background or career interests.

Problem Statement

Develop an AI-enabled placement-readiness platform that analyses student profiles, resumes, coding performance, aptitude results and preferred job roles. The system should identify skill gaps and generate a personalized preparation roadmap. It should recommend learning resources, practice assessments and suitable job categories. Faculty placement coordinators should be able to monitor batch-level readiness and identify common skill deficiencies. The platform should track progress and update recommendations after every assessment.

Mandatory Requirements

- Student-profile and resume analysis
- Role-based skill-gap identification
- Personalized preparation roadmap
- Assessment and progress tracking
- Suitable-role recommendations
- Placement-coordinator analytics

Expected Prototype

A functional platform demonstrating profile analysis, skill-gap reporting, learning recommendations and placement-readiness tracking.

KH26-CSE-10 – AI-Assisted Cybersecurity Incident Prioritization and Response Platform

Background

Institutions receive many security alerts related to suspicious logins, phishing attempts, unusual file access and abnormal network activity. Reviewing every alert manually can delay the response to genuinely critical incidents.

Problem Statement

Develop a defensive cybersecurity platform that receives security-event records and categorizes them according to severity and possible impact. The system should correlate related alerts, identify unusual activity and recommend safe response actions to administrators. It should display an incident timeline, affected systems and the reason for assigning a particular risk level. Administrators must be able to acknowledge, investigate, escalate and close incidents. The solution should focus on authorized monitoring and defensive security only.

Mandatory Requirements

- Security-event ingestion
- AI-based anomaly and severity analysis
- Related-alert correlation
- Explainable risk scoring
- Incident response workflow
- Security analytics dashboard

Expected Prototype

A defensive monitoring dashboard demonstrating alert analysis, incident prioritization and response tracking using simulated security logs.